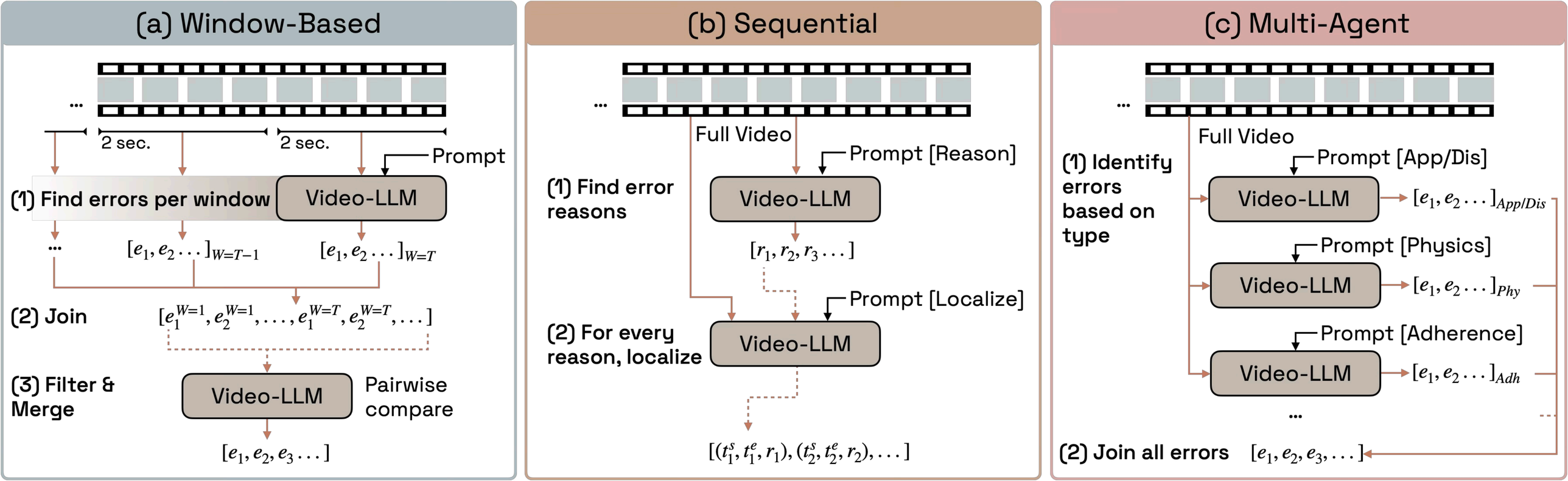

(a) Window-Based
2 sec.
2 sec.
Prompt
(1) Find errors per window
Video-LLM
$[e_1, e_2 \dots]_{W=T-1}$
$[e_1, e_2 \dots]_{W=T}$
(2) Join
$[e_1^{W=1}, e_2^{W=1}, \dots, e_1^{W=T}, e_2^{W=T}, \dots]$
(3) Filter & Merge
Video-LLM
Pairwise compare
$[e_1, e_2, e_3 \dots]$
(b) Sequential
Full Video
Prompt [Reason]
(1) Find error reasons
Video-LLM
$[r_1, r_2, r_3 \dots]$
Prompt [Localize]
(2) For every reason, localize
Video-LLM
$[(t_1^s, t_1^e, r_1), (t_2^s, t_2^e, r_2), \dots]$
(c) Multi-Agent
Full Video
(1) Identify errors based on type
Prompt [App/Dis]
Video-LLM
$[e_1, e_2 \dots]_{App/Dis}$
Prompt [Physics]
Video-LLM
$[e_1, e_2 \dots]_{Phy}$
Prompt [Adherence]
Video-LLM
$[e_1, e_2 \dots]_{Adh}$
(2) Join all errors
$[e_1, e_2, e_3, \dots]$